

STRENGTH OF MATERIALS

Complex Stress

Stress on Inclined Plane/ Stress Transformation/

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OBJECTIVES

At the end of this lesson students will be able to:

- Evaluate stresses on any plane through stress-transformation equations.
- Evaluate principal stresses and locate principal planes for plane stress state
- Find maximum shear stresses at a point on a body for plane stress state
- Construct Mohr's circle for stress and identify the points of principal stress and maximum shear stresses

KEY TERMS

- Plane stress
- Complex stress
- Stress transformation
- Stress transformation equations
- Principal stress
- Principal plane

1.1 Introduction

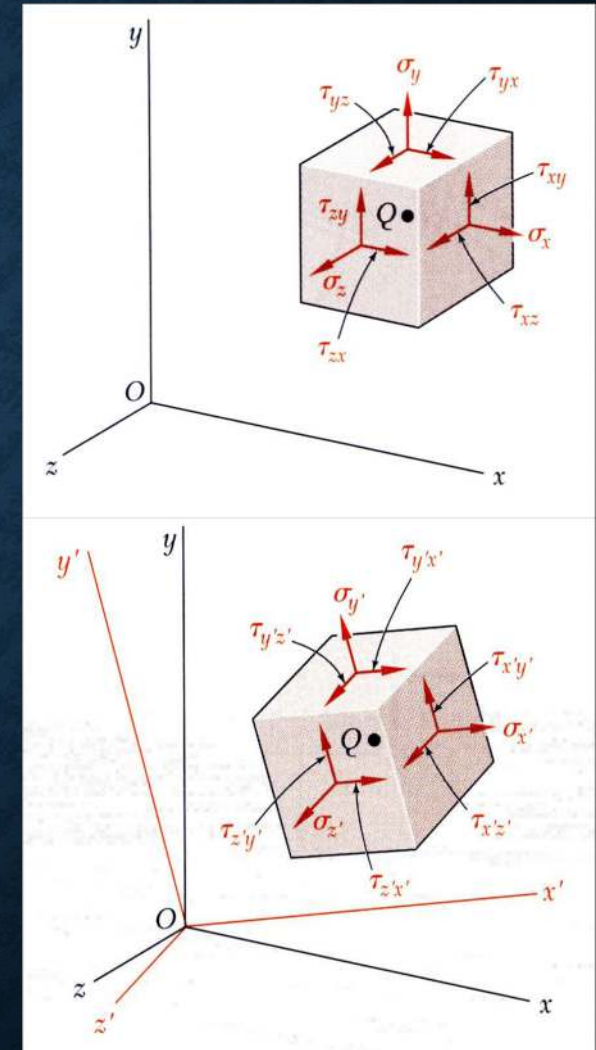
General state of stress:

➤ The most general state of stress *at a given point Q* may be represented by 6 components

- Three Normal stresses ($\sigma_x, \sigma_y, \sigma_z$) and
- Three Shearing stresses ($\tau_{xy}, \tau_{yz}, \tau_{zx}$)

(Note that $\tau_{xy} = \tau_{yx}, \tau_{yz} = \tau_{zy}, \tau_{zx} = \tau_{xz}$)

➤ Same state of stress is represented by a different set of components if axes are rotated



PLANE STRESS

- This section is concerned with a special two-dimensional state of stress called **plane stress**.
- It is important for two reasons:
 - (1) It arises in real components (particularly in thin components loaded in certain ways), and
 - (2) It is a two dimensional state of stress, and thus serves as an excellent introduction to more complicated three dimensional stress states.

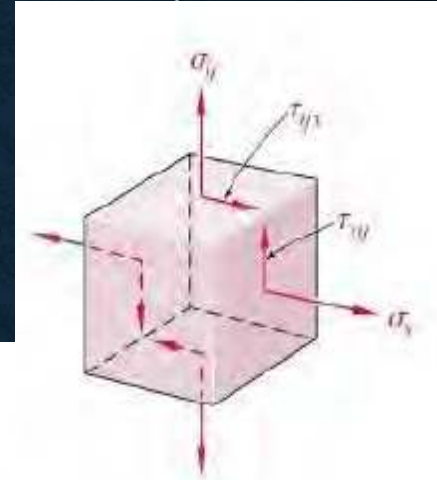
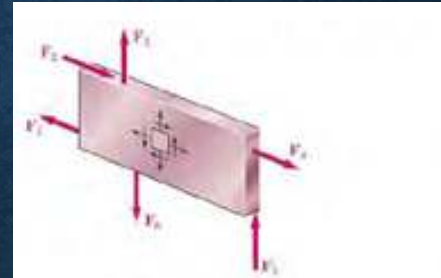
Plane Stress:

➤ A situation in which two of the faces of the cubic element are free of any stress

- For the previous general state of stress case,
the state of plane stress is defined by σ_x , σ_y , τ_{xy} and

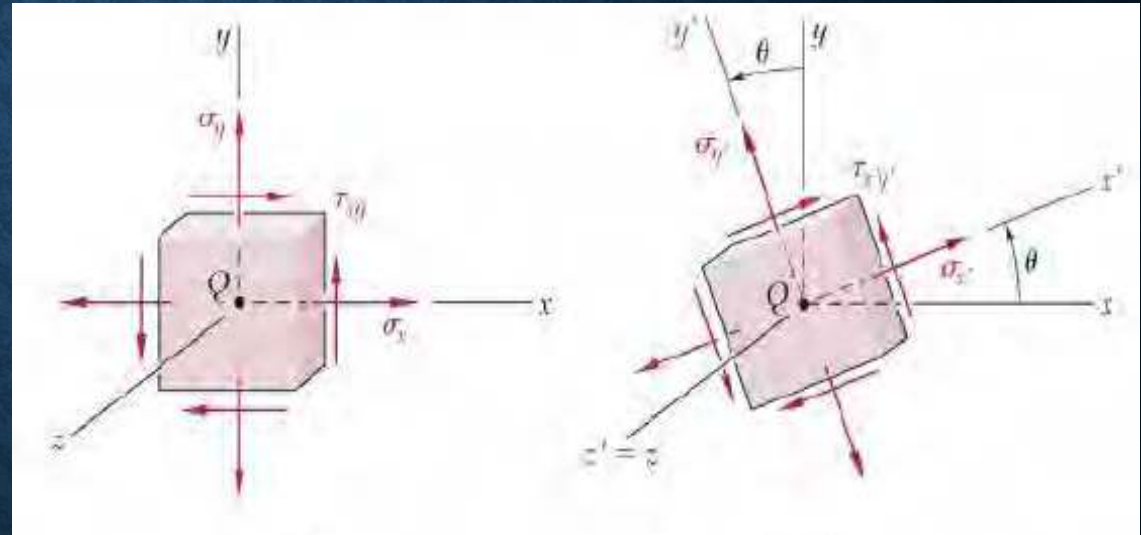
$$\sigma_z = \tau_{zx} = \tau_{zy} = 0$$

- State of plane stress occurs :
 - in a thin plate subjected to forces acting in the mid plane of the plate and
 - on the free surface of a structural element or machine component, i.e., at any point of the surface not subjected to an external force



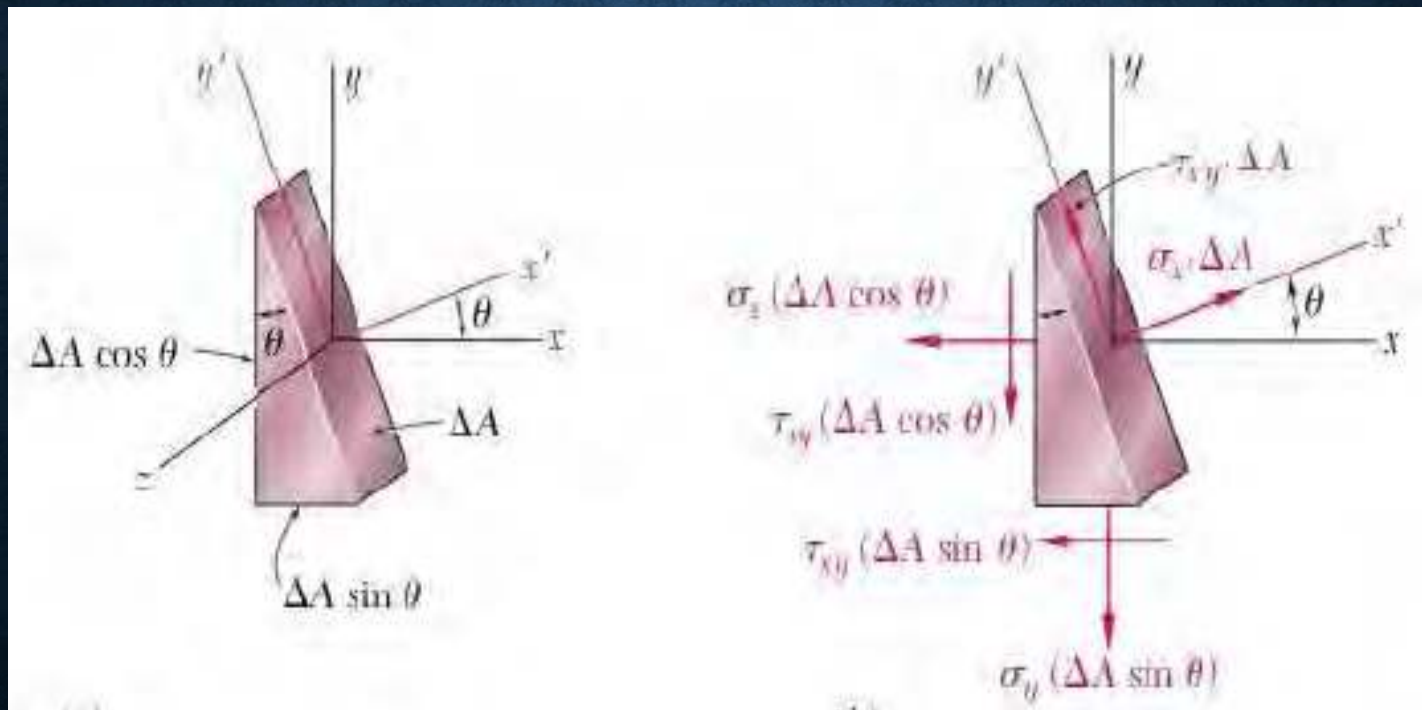
1.2 Stresses on Oblique Plane (Stress Transformation)

- Consider a state of plane stress exists at point Q (with $\sigma_z = \tau_{zx} = \tau_{zy} = 0$) and it is characterized by the stress components σ_x , σ_y , and τ_{xy} associated with the element shown below (left figure).



- The stress components $\sigma_{x'}$, $\sigma_{y'}$, and $\tau_{x'y'}$ associated with the element, after it has been rotated through an angle θ about the z axis (right figure), can be expressed in terms of σ_x , σ_y , τ_{xy} and θ using stress transformation equation.

- To determine the normal stress $\sigma_{x'}$ and the shear stress $\tau_{x'y'}$ exerted on the face perpendicular to the x' axis, let's consider a prismatic element with faces respectively perpendicular to x , y and x' axes.



➤ Using the equilibrium equation along the x' and y' axes

$$\begin{aligned}\sum F_{x'} = 0 = & \sigma_{x'} \Delta A - \sigma_x (\Delta A \cos \theta) \cos \theta - \tau_{xy} (\Delta A \cos \theta) \sin \theta \\ & - \sigma_y (\Delta A \sin \theta) \sin \theta - \tau_{xy} (\Delta A \sin \theta) \cos \theta\end{aligned}$$

$$\begin{aligned}\sum F_{y'} = 0 = & \tau_{x'y'} \Delta A + \sigma_x (\Delta A \cos \theta) \sin \theta - \tau_{xy} (\Delta A \cos \theta) \cos \theta \\ & - \sigma_y (\Delta A \sin \theta) \cos \theta + \tau_{xy} (\Delta A \sin \theta) \sin \theta\end{aligned}$$

➤ Solving the first equation for $\sigma_{x'}$, and the second for $\tau_{x'y'}$, we have

$$\sigma_{x'} = \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + 2\tau_{xy} \sin \theta \cos \theta \dots\dots\dots \text{(iii)}$$

$$\tau_{x'y'} = -(\sigma_x - \sigma_y) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta) \dots\dots\dots \text{(iv)}$$

➤ Recalling the trigonometric relations

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta & \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ \text{and} \\ \cos^2 \theta &= \frac{1 + \cos 2\theta}{2} & \sin^2 \theta &= \frac{1 - \cos 2\theta}{2} \end{aligned}$$

and rearranging we write eqns. (iii) and (iv) as follows

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \dots\dots\dots (1.1)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \dots\dots\dots (1.2)$$

❖ These equations are usually called the **transformation equations for plane stress** because they transform the stress components from one set of axes **(x, y)** to another **(x', y')**

- The expression for the normal stress $\sigma_{y'}$, is obtained by replacing θ in Eq. (1.1) by the angle $\theta + 90^\circ$ that the y' axis forms with the x axis.

Since $\cos (2\theta + 180^\circ) = -\cos 2\theta$ and $\sin (2\theta + 180^\circ) = -\sin 2\theta$, we have

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta \dots\dots\dots (1.3)$$

- Adding Eqs. (1.1) and (1.3) member to member, we obtain

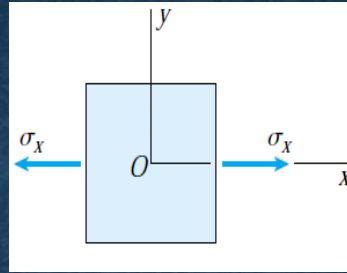
$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y \dots\dots\dots (1.4)$$

- ❖ Thus, in the case of plane stress the sum of the normal stresses exerted on a cubic element of material is independent of the orientation of that element.

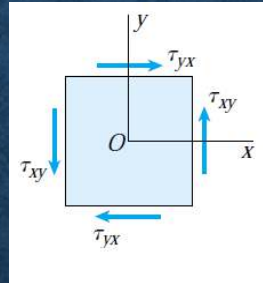
Special Case of Plane Stress

I) Uni-Axial State

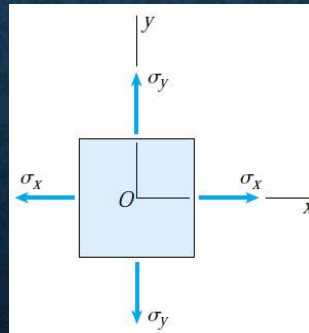
$$\sigma_{x'} = \sigma_x \cos^2 \theta$$



II) Pure Shear



III) Bi-Axial State (Pure Normal)



1.3 Principal Stresses and Maximum Shear Stress

- The transformation equations for plane stress (eqns. (1.1) and (1.2)) show that the normal stresses $\sigma_{x'}$ and the shear stresses $\tau_{x'y'}$ vary continuously as the axes are rotated through the angle θ .
- Both the normal and shear stresses reach maximum and minimum values within a given interval and these maximum and minimum values are usually needed for design purposes.

Principal Stresses

- The maximum and minimum normal stresses, called the **principal stresses**, can be found from the transformation equation for the normal stress $\sigma_{x'}$ (Eq. 1.1), by taking the derivative of $\sigma_{x'}$ with respect to θ and setting it equal to zero, we obtain an equation from which we can find the values of θ at which $\sigma_{x'}$ is a maximum or a minimum.

➤ Using Eqn. 1.1 and taking its first derivative with respect to θ

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\frac{d\sigma_{x'}}{d\theta} = -(\sigma_x - \sigma_y) \sin 2\theta + 2\tau_{xy} \cos 2\theta = 0$$

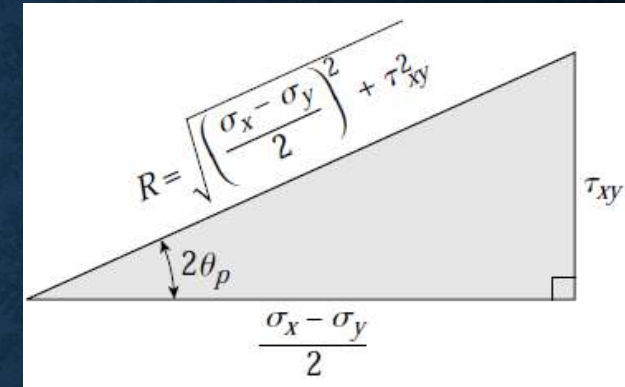
➤ From which we get the principal plane

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \dots\dots\dots (1.5)$$

- The subscript p indicates that the angle θ_p defines the orientation of the **principal planes**, that is, the planes on which the principal stresses act.

- Let's refer to the right triangle in the figure below, which is constructed from Eq. (1.5). Note that the hypotenuse of the triangle, obtained from the Pythagorean theorem, is

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



- From the triangle we obtain two additional relations:

$$\cos 2\theta_p = \frac{\sigma_x - \sigma_y}{2R} \quad \sin 2\theta_p = \frac{\tau_{xy}}{R}$$

- Substituting these expressions for $\cos 2\theta$ and $\sin 2\theta$ into Eq. (1.1) and obtain the algebraically larger of the two principal stresses, denoted by σ_1 :

$$\begin{aligned} \sigma_1 = \sigma_{x_1} &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta_p + \tau_{xy} \sin 2\theta_p \\ &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \left(\frac{\sigma_x - \sigma_y}{2R} \right) + \tau_{xy} \left(\frac{\tau_{xy}}{R} \right) \end{aligned}$$

➤ After substituting for R and performing some algebraic manipulations, we obtain

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (v)$$

➤ The smaller of the principal stresses, denoted by σ_2 , may be found from the condition that the sum of the normal stresses on perpendicular planes is constant (see Eq. 1.4):

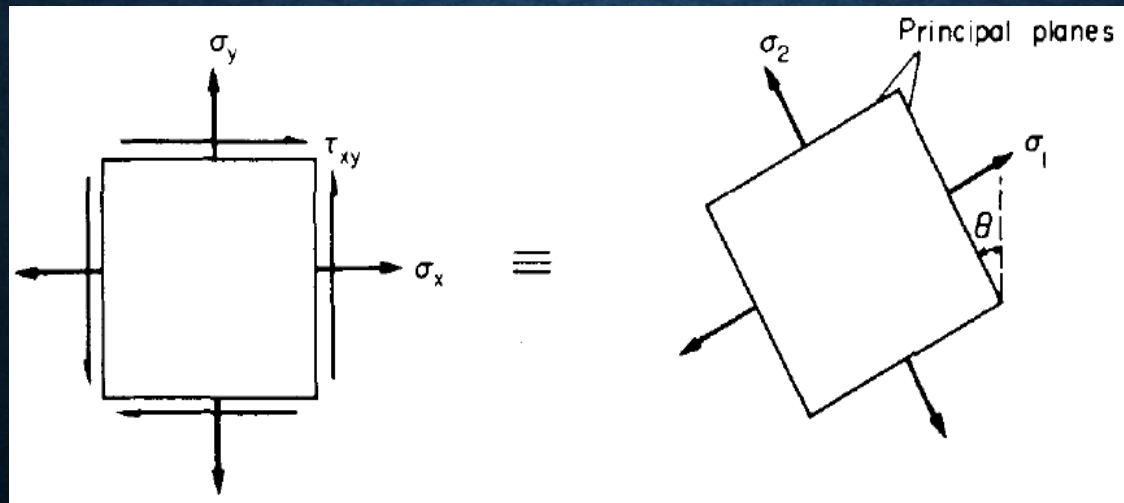
$$\sigma_1 + \sigma_2 = \sigma_x + \sigma_y \dots\dots\dots (1.4)$$

➤ Substituting the expression for σ_1 into Eq. (1.4) and solving for σ_2 , we get

$$\begin{aligned} \sigma_2 &= \sigma_x + \sigma_y - \sigma_1 \\ &= \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (vi) \end{aligned}$$

➤ Combining Eqns. (v) and (vi) in to a single formula the famous *Principal Stresses* formula can be expressed as

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



Maximum Shear Stress

- The maximum Shear Stress can be found from the transformation equation for the shear stress $\tau_{x'y'}$ (Eq. 1.2), by taking the derivative of $\tau_{x'y'}$ with respect to θ and setting it equal to zero, we obtain an equation from which we can find the values of θ at which $\tau_{x'y'}$ is a maximum.
- Using Eqn. 1.2 and taking its first derivative with respect to θ

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

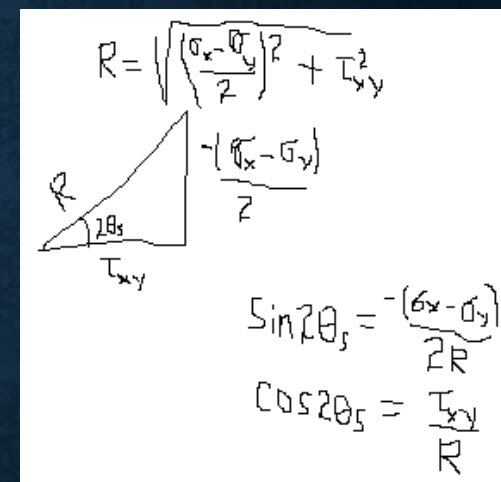
$$\frac{d\tau_{x'y'}}{d\theta} = -(\sigma_x - \sigma_y) \cos 2\theta - 2\tau_{xy} \sin 2\theta = 0$$

- From which we get the plane at which the shear stress is maximum

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}} \dots\dots\dots (1.7)$$

- The subscript s indicates that the angle θ_s defines the orientation of the maximum shear **plane**, that is, the planes on which the maximum shear stress acts.
- From Eqn. (1.7) we can construct a right triangle as shown in the figure below.
- Substituting these expressions for $\cos 2\theta$ and $\sin 2\theta$ into Eq. (1.2), the maximum shear stress can be obtained.

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (1.8)$$



- Equating Eqn. (1.5) with Eqn. (1.7) we will get a relationship between principal plane θ_p and maximum shear plane θ_s .

$$\theta_s = \theta_p - 45^\circ \dots\dots\dots(1.9)$$

1.4 Mohr's Circle of Stress

- The transformation equations for plane stress can be represented in graphical form by a plot known as **Mohr's circle**.
- **Mohr's Circle of Stress:**
 - Enables to visualize the relationships between the normal and shear stresses acting on various inclined planes at a point in a stressed body
 - Provides a means for calculating principal stresses, maximum shear stresses, and stresses on inclined planes

Equations of Mohr's Circle

- Can be derived from the two transformation equations for plane stress, (Eqns. (1) and (2)), with a slight rearrangement of the first equation.

$$\sigma_{x'} - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

..... (vii)

- Squaring both sides of each equation and then adding the two equations to eliminate the parameter 2θ ,

$$\left(\sigma_{x'} - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{x'y'}^2 = \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2 \dots\dots\dots (viii)$$

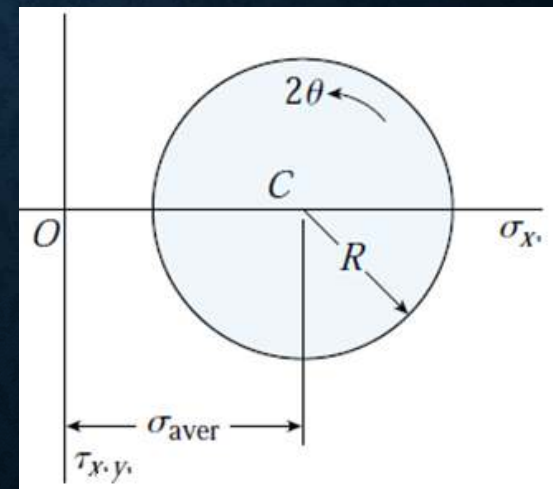
- Representing *some terms* by σ_{aver} and R ,

$$\sigma_{aver} = \frac{\sigma_x + \sigma_y}{2} \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \dots\dots\dots (ix)$$

the equation of the Mohr's Circle can be written as

$$(\sigma_{x'} - \sigma_{aver})^2 + \tau_{x'y'}^2 = R^2 \dots\dots\dots (1.10)$$

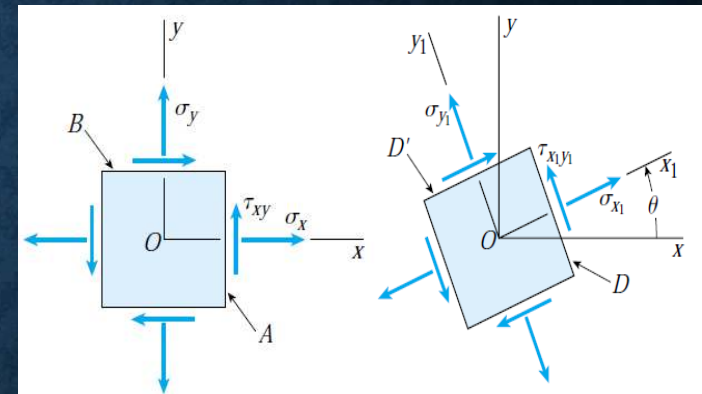
Which is the equation of a circle,
with horizontal axis σ' and vertical axis $\tau_{x'y'}$,
and its center $(0, \sigma_{aver})$ and radius R



Construction of Mohr's Circle

➤ With σ_x , σ_y and τ_{xy} known, the procedure for constructing Mohr's circle is as follows

1. Draw a set of coordinate axes with $\sigma_{x'}$ as abscissa (positive to the right) and $\tau_{x'y'}$ as ordinate (positive downward)
2. Locate the center C of the circle at the point having coordinates $\sigma_{x'} = \sigma_{ave}$ and $\tau_{x'y'} = 0$
3. Locate point A , representing the stress conditions on the x face of the element shown in Fig. 7-15a, by plotting its coordinates $\sigma_{x'} = \sigma_x$ and $\tau_{x'y'} = \tau_{xy}$.



Note that point A on the circle corresponds to $\theta = 0$.

Also, note that the x face of the element (Fig. 7-15a) is labeled “ A ” to show its correspondence with point A on the circle.

4. Locate point B , representing the stress conditions on the y face of the element shown in Fig. 7-15a, by plotting its coordinates $\sigma_{x'} = \sigma_y$ and $\tau_{x'y'} = -\tau_{xy}$. Note that point B on the circle corresponds to $\theta = 90^\circ$. In addition, the y face of the element (Fig. 7-15a) is labeled “ B ” to show its correspondence with point B on the circle.

5. Draw a line from point A to point B . This line is a diameter of the circle and passes through the center C . Points A and B , representing the stresses on planes at 90° to each other (Fig. 7-15a), are at opposite ends of the diameter (and therefore are 180° apart on the circle).

6. Using point C as the center, draw Mohr's circle through points A and B . The circle drawn in this manner has radius R

